Homework Assignment 2: Exploring patterns of environmental justice

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Part 1: Legacy of redlining in current environmental (in)justice

Import packages and read the datasets

```
library(tmap)
```

```
Registered S3 method overwritten by 'stars':
  method                   from
  st_interpolate_aw.stars sf
```

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2

-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```r
library(sf)
```

```
Linking to GEOS 3.14.1, GDAL 3.12.1, PROJ 9.7.1; sf_use_s2() is TRUE
```

```r
# Home Owners' Loan Corporation redlining dataset
holc <- st_read("data/mapping-inequality-los-angeles.json")
```

```
Reading layer `mapping-inequality-los-angeles' from data source
  `C:\Users\aaron\Documents\Git\eds223-assignment-2\data\mapping-inequality-
los-angeles.json'
  using driver `GeoJSON'
Simple feature collection with 417 features and 14 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -118.6104 ymin: 33.70563 xmax: -117.7028 ymax: 34.30388
Geodetic CRS:  WGS 84
```

```r
# Statewide Integrated Traffice Records System transportation accidents from 2020 - 2025
crash <- st_read("data/Crashes.csv")
```

```
Reading layer `Crashes' from data source
  `C:\Users\aaron\Documents\Git\eds223-assignment-2\data\Crashes.csv'
  using driver `CSV'

Warning: no simple feature geometries present: returning a data.frame or tbl_df
```

```r
ejscreen <- sf::st_read(here::here("data","EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb")
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
  `C:\Users\aaron\Documents\Git\eds223-assignment-2\data\EJSCREEN_2023_BG_StatePct_with_AS_CN
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS: WGS 84 / Pseudo-Mercator
```

```r
# filter to a state you are interested in
california <- ejscreen %>%
  dplyr::filter(ST_ABBREV == "CA")

# filter to a county you are interested in
los_angeles <- ejscreen %>%
  dplyr::filter(CNTY_NAME %in% c("Los Angeles County"))
```

Dataset Testing/Exploration Pane

```r
# observe the coordinate reference system for each dataset
st_crs(holc)
```

```
Coordinate Reference System:
  User input: WGS 84
  wkt:
GEOGCRS["WGS 84",
    DATUM["World Geodetic System 1984",
        ELLIPSOID["WGS 84",6378137,298.257223563,
            LENGTHUNIT["metre",1]]],
    PRIMEM["Greenwich",0,
        ANGLEUNIT["degree",0.0174532925199433]],
    CS[ellipsoidal,2],
        AXIS["geodetic latitude (Lat)",north,
            ORDER[1],
            ANGLEUNIT["degree",0.0174532925199433]],
        AXIS["geodetic longitude (Lon)",east,
            ORDER[2],
            ANGLEUNIT["degree",0.0174532925199433]],
    ID["EPSG",4326]]
```

```r
st_crs(los_angeles)
```

```
Coordinate Reference System:
  User input: WGS 84 / Pseudo-Mercator
  wkt:
PROJCRS["WGS 84 / Pseudo-Mercator",
    BASEGEOGCRS["WGS 84",
        ENSEMBLE["World Geodetic System 1984 ensemble",
```

```
        MEMBER["World Geodetic System 1984 (Transit)"],
        MEMBER["World Geodetic System 1984 (G730)"],
        MEMBER["World Geodetic System 1984 (G873)"],
        MEMBER["World Geodetic System 1984 (G1150)"],
        MEMBER["World Geodetic System 1984 (G1674)"],
        MEMBER["World Geodetic System 1984 (G1762)"],
        MEMBER["World Geodetic System 1984 (G2139)"],
        MEMBER["World Geodetic System 1984 (G2296)"],
        ELLIPSOID["WGS 84",6378137,298.257223563,
            LENGTHUNIT["metre",1]],
        ENSEMBLEACCURACY[2.0]],
    PRIMEM["Greenwich",0,
        ANGLEUNIT["degree",0.0174532925199433]],
    ID["EPSG",4326]],
CONVERSION["Popular Visualisation Pseudo-Mercator",
    METHOD["Popular Visualisation Pseudo Mercator",
        ID["EPSG",1024]],
    PARAMETER["Latitude of natural origin",0,
        ANGLEUNIT["degree",0.0174532925199433],
        ID["EPSG",8801]],
    PARAMETER["Longitude of natural origin",0,
        ANGLEUNIT["degree",0.0174532925199433],
        ID["EPSG",8802]],
    PARAMETER["False easting",0,
        LENGTHUNIT["metre",1],
        ID["EPSG",8806]],
    PARAMETER["False northing",0,
        LENGTHUNIT["metre",1],
        ID["EPSG",8807]]],
CS[Cartesian,2],
    AXIS["easting (X)",east,
        ORDER[1],
        LENGTHUNIT["metre",1]],
    AXIS["northing (Y)",north,
        ORDER[2],
        LENGTHUNIT["metre",1]],
USAGE[
    SCOPE["Web mapping and visualisation."],
    AREA["World between 85.06°S and 85.06°N."],
    BBOX[-85.06,-180,85.06,180]],
ID["EPSG",3857]]
```

```r
# check the first couple rows for each column
head(crash)
```

```
  CASE_ID ACCIDENT_YEAR  PROC_DATE JURIS COLLISION_DATE COLLISION_TIME
1 8008484          2020 2020-11-20  1942     2020-08-06           2125
2 8008488          2020 2020-10-23  1942     2020-08-06           1920
3 8008496          2020 2020-10-24  1942     2020-08-13           1745
4 8008508          2020 2020-12-04  1942     2020-09-02           1220
5 8008516          2020 2020-12-01  1942     2020-09-11           1115
6 8008558          2020 2020-12-19  1942     2020-10-02           0725
  OFFICER_ID REPORTING_DISTRICT DAY_OF_WEEK CHP_SHIFT POPULATION CNTY_CITY_LOC
1      18124              1494           4         5          7          1942
2      41957              1434           4         5          7          1942
3      32820              1405           4         5          7          1942
4      34209                             3         5          7          1942
5      33869              0806           5         5          7          1942
6      44435              0514           5         5          7          1942
  SPECIAL_COND BEAT_TYPE CHP_BEAT_TYPE CITY_DIVISION_LAPD CHP_BEAT_CLASS
1            0         0             0                  N              0
2            0         0             0                  N              0
3            0         0             0                  N              0
4            0         0             0                                 0
5            0         0             0                  H              0
6            0         0             0                  E              0
  BEAT_NUMBER   PRIMARY_RD     SECONDARY_RD DISTANCE DIRECTION INTERSECTION
1       3A101 IMPERIAL HWY          MAIN ST    33.00         E            N
2       14A25    VENICE BL   S CENTINELA AV    40.00         W            N
3       14A56  NATIONAL BL CHEVIOT VISTA PL      .00                      Y
4       14A49   WASHINGTON          DELL AV      .00         E            N
5         8A1   BEL AIR RD        CUESTA WY      .00                      Y
6        5A25   ANAHEIM ST    WILMINGTON BL      .00                      Y
  WEATHER_1 WEATHER_2 STATE_HWY_IND CALTRANS_COUNTY CALTRANS_DISTRICT
1         A         -            N
2         A         -            N
3         A         -            N
4         A         -            N
5         A         -            N
6         A         -            N
  STATE_ROUTE ROUTE_SUFFIX POSTMILE_PREFIX POSTMILE LOCATION_TYPE
1
2
3
```

4
5
6

	RAMP_INTERSECTION	SIDE_OF_HWY	TOW_AWAY	COLLISION_SEVERITY	NUMBER_KILLED
1			Y	4	0
2			N	4	0
3			Y	3	0
4			N	4	0
5			Y	4	0
6			Y	4	0

	NUMBER_INJURED	PARTY_COUNT	PRIMARY_COLL_FACTOR	PCF_CODE_OF_VIOL
1	2	2	A	-
2	2	2	A	-
3	2	3	A	-
4	1	2	A	-
5	1	2	A	-
6	3	3	A	-

	PCF_VIOL_CATEGORY	PCF_VIOLATION	PCF_VIOL_SUBSECTION	HIT_AND_RUN
1	03	22350		N
2	03	22350		N
3	08	22107		N
4	05	21650	A	N
5	03	22350		N
6	09	21801		N

	TYPE_OF_COLLISION	MVIW	PED_ACTION	ROAD_SURFACE	ROAD_COND_1	ROAD_COND_2
1	B	C	A	A	H	-
2	C	C	A	A	H	-
3	B	C	A	A	H	-
4	B	C	A	A	H	-
5	A	C	A	A	H	-
6	A	C	A	A	H	-

	LIGHTING	CONTROL_DEVICE	CHP_ROAD_TYPE	PEDESTRIAN_ACCIDENT	BICYCLE_ACCIDENT
1	C	A	0		
2	A	A	0		
3	A	A	0		
4	A	D	0		
5	A	D	0		
6	A	A	0		

	MOTORCYCLE_ACCIDENT	TRUCK_ACCIDENT	NOT_PRIVATE_PROPERTY	ALCOHOL_INVOLVED
1		Y	Y	
2			Y	
3			Y	
4		Y	Y	

```
5                                                    Y
6                                                    Y
  STWD_VEHTYPE_AT_FAULT CHP_VEHTYPE_AT_FAULT COUNT_SEVERE_INJ COUNT_VISIBLE_INJ
1                     F                   26                0                 0
2                     D                   22                0                 0
3                     D                   22                0                 1
4                     -                    -                0                 0
5                     D                   22                0                 0
6                     A                   07                0                 0
  COUNT_COMPLAINT_PAIN COUNT_PED_KILLED COUNT_PED_INJURED
1                    2                0                 0
2                    2                0                 0
3                    1                0                 0
4                    1                0                 0
5                    1                0                 0
6                    3                0                 0
  COUNT_BICYCLIST_KILLED COUNT_BICYCLIST_INJURED COUNT_MC_KILLED
1                      0                       0               0
2                      0                       0               0
3                      0                       0               0
4                      0                       0               0
5                      0                       0               0
6                      0                       0               0
  COUNT_MC_INJURED PRIMARY_RAMP SECONDARY_RAMP LATITUDE LONGITUDE      COUNTY
1                0            -              -                    LOS ANGELES
2                0            -              -                    LOS ANGELES
3                0            -              -                    LOS ANGELES
4                0            -              -                    LOS ANGELES
5                0            -              -                    LOS ANGELES
6                0            -              -                    LOS ANGELES
         CITY             POINT_X            POINT_Y
1 LOS ANGELES -118.41586303710938 33.931461334228516
2 LOS ANGELES  -118.4337158203125 34.003692626953125
3 LOS ANGELES  -118.4146728515625 34.030849456787109
4 LOS ANGELES -118.46352386474609 33.981048583984375
5 LOS ANGELES -118.44111633300781 34.085971832275391
6 LOS ANGELES   -118.274169921875 33.779331207275391
```

Transforming coordinate reference system to match

```
holc <- st_transform(holc, crs = st_crs(los_angeles))
```

Coordinate Reference System Verification

```
if (st_crs(holc) == st_crs(los_angeles)) {
  print('crs matches')
}else{
  print('crs do not match')
}
```

```
[1] "crs matches"
```

```
if(st_crs(crash) == st_crs(los_angeles)) {
  print('crs matches')
} else {
  warning('crs do not match!')
}
```

```
Warning: crs do not match!
```

1. Create a map of historical redlining neighborhoods, including:

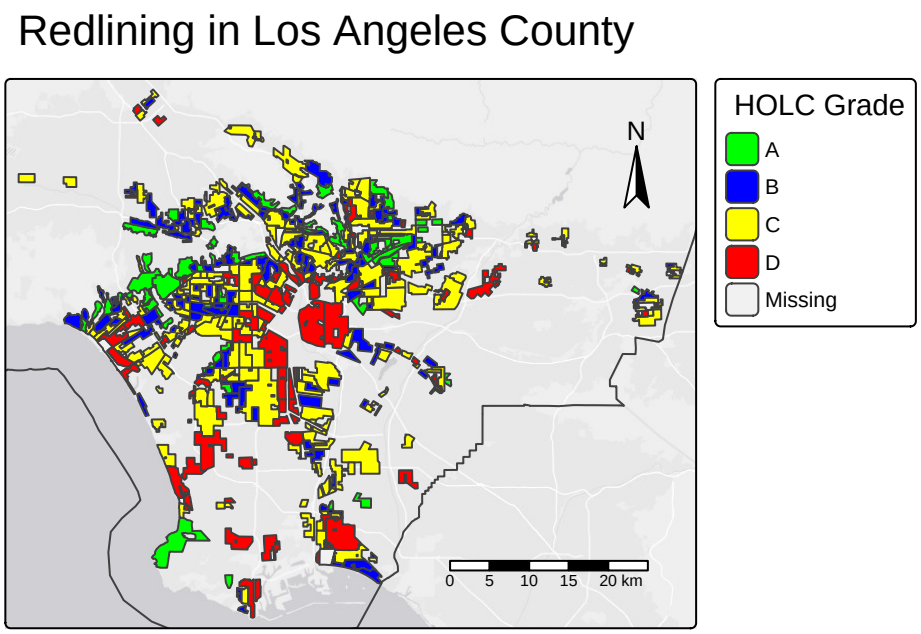
- neighborhoods colored by HOLC grade
- an appropriate base map
- a clear title and legend

```
# graph HOLC grade in each census block neighborhood
la_boundary <- st_union(los_angeles)
tm_shape(holc) +
    tm_polygons("grade",
                fill.scale = tm_scale_categorical(values = c(
                    "green",
                    "blue",
                    "yellow",
                    "red"
                )),
                fill.legend = tm_legend(title = " HOLC Grade")
```

```
    ) +

# add county boundary and basemap with scale orientation
tm_shape(la_boundary) +
    tm_borders() +
tm_title(text = "Redlining in Los Angeles County") +
tm_compass(position = c('top', 'right')) +
tm_scalebar(position = c('bottom', 'right'), size = 1) +
tm_basemap(server = "Esri.WorldGrayCanvas")
```

```
! The 'size' argument of `tm_scalebar()` is deprecated as of tmap 4.0.
i Please use 'text.size' instead.
Multiple palettes called "green" found: "kovesi.green", "tableau.green". The first one, "kove
```



2. Create a table summarizing:

- the percentage of census block groups that fall within each HOLC grade
- Also include the percent of census black groups that don't fall within a HOLC grade in your table

- Hint: The HOLC data contains the grades and the EJScreen data contains the census blocks, so you will need to combine the data spatially before doing summary statistics. Once you combine and no longer need the geometries, you can use st_drop_geometry().

```
# join los angeles and HOLC datasets
la_holc <- st_join(los_angeles, holc)

# summarize percentage of census block groups within each grade
holc_table <- la_holc |>
  st_drop_geometry() |>
  drop_na(grade) |>
  group_by(grade) |>
  summarize(n = n()) |>
  mutate(percentage_census = n / sum(n) * 100)
```

3. Create at least two visualizations summarizing current conditions (from the EJScreen data) within HOLC grades using the mean of the following variables (you may combine variables in a single visualization or look at a single variable per plot):

- % low income
- percentile for Particulate Matter 2.5
- percentile for low life expectancy
- Use ggplot for your visualizations! You will first need to calculate mean of each variable grouped by HOLC grade.

```
# calculate the average value of each variable in each grade
summarize_variables <- la_holc |>
  st_drop_geometry() |>
  drop_na(grade) |>
  group_by(grade) |>
  summarize(low_income_mean = mean(LOWINCPCT, na.rm = TRUE),
            pm25_mean = mean(P_PM25, na.rm = TRUE),
            life_expectancy_mean = mean(P_LIFEEXPPCT, na.rm = TRUE)
  )

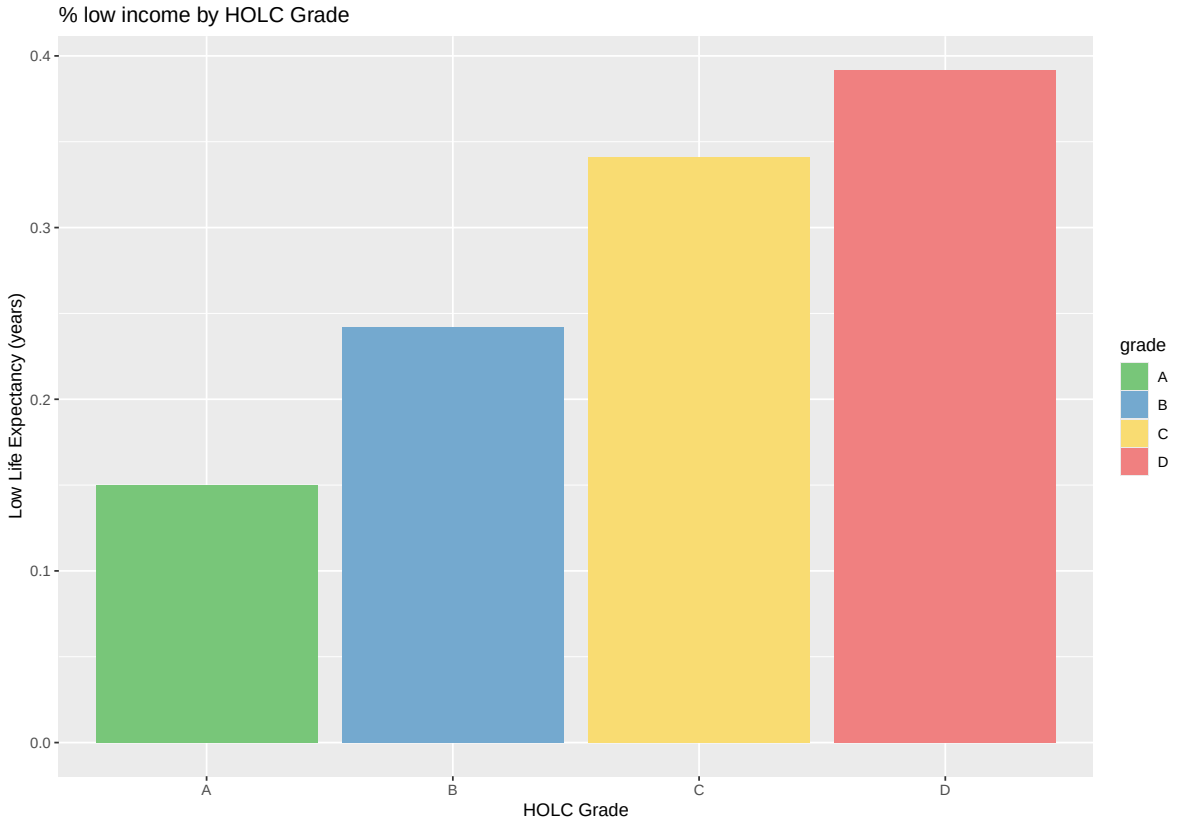
# create barplots comparing average low-income percentage, PM2.5 exposure, and low life expe
p1 <- ggplot(summarize_variables, aes(x = grade, y = low_income_mean, fill = grade)) +
  geom_col() +
  scale_fill_manual(values = c(
          "A" = "#78C679",
          "B" = "#74A9CF",
```

```
          "C" = "#F9DC72",
          "D" = "#F08080"
  )) +
  labs(
    title = "% low income by HOLC Grade",
    x = "HOLC Grade",
    y = "Low Life Expectancy (years)"
  )

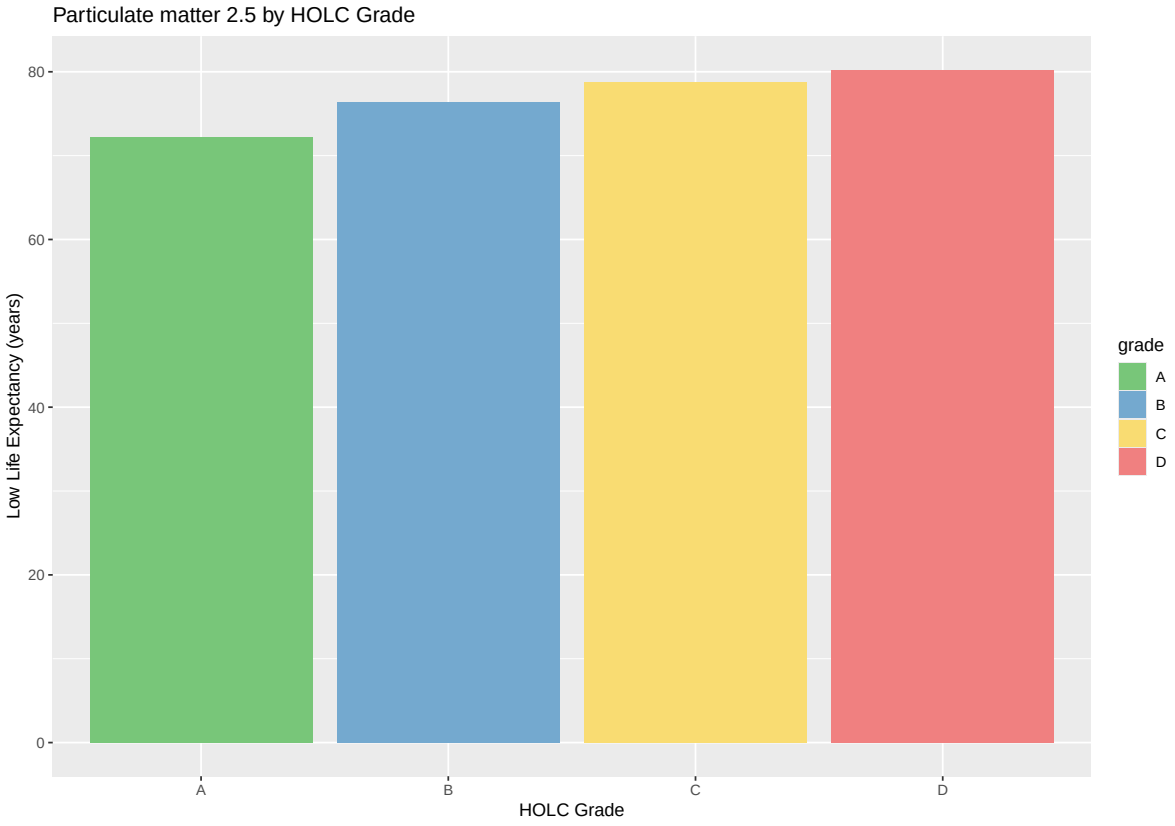
p2 <- ggplot(summarize_variables, aes(x = grade, y = pm25_mean, fill = grade)) +
  geom_col() +
  scale_fill_manual(values = c(
          "A" = "#78C679",
          "B" = "#74A9CF",
          "C" = "#F9DC72",
          "D" = "#F08080"
  )) +
  labs(
    title = "Particulate matter 2.5 by HOLC Grade",
    x = "HOLC Grade",
    y = "Low Life Expectancy (years)"
  )

p3 <- ggplot(summarize_variables, aes(x = grade, y = life_expectancy_mean, fill = grade)) +
  geom_col() +
  scale_fill_manual(values = c(
          "A" = "#78C679",
          "B" = "#74A9CF",
          "C" = "#F9DC72",
          "D" = "#F08080"
  )) +
  labs(
    title = "Low Life Expectancy by HOLC Grade",
    x = "HOLC Grade",
    y = "Low Life Expectancy (years)"
  )

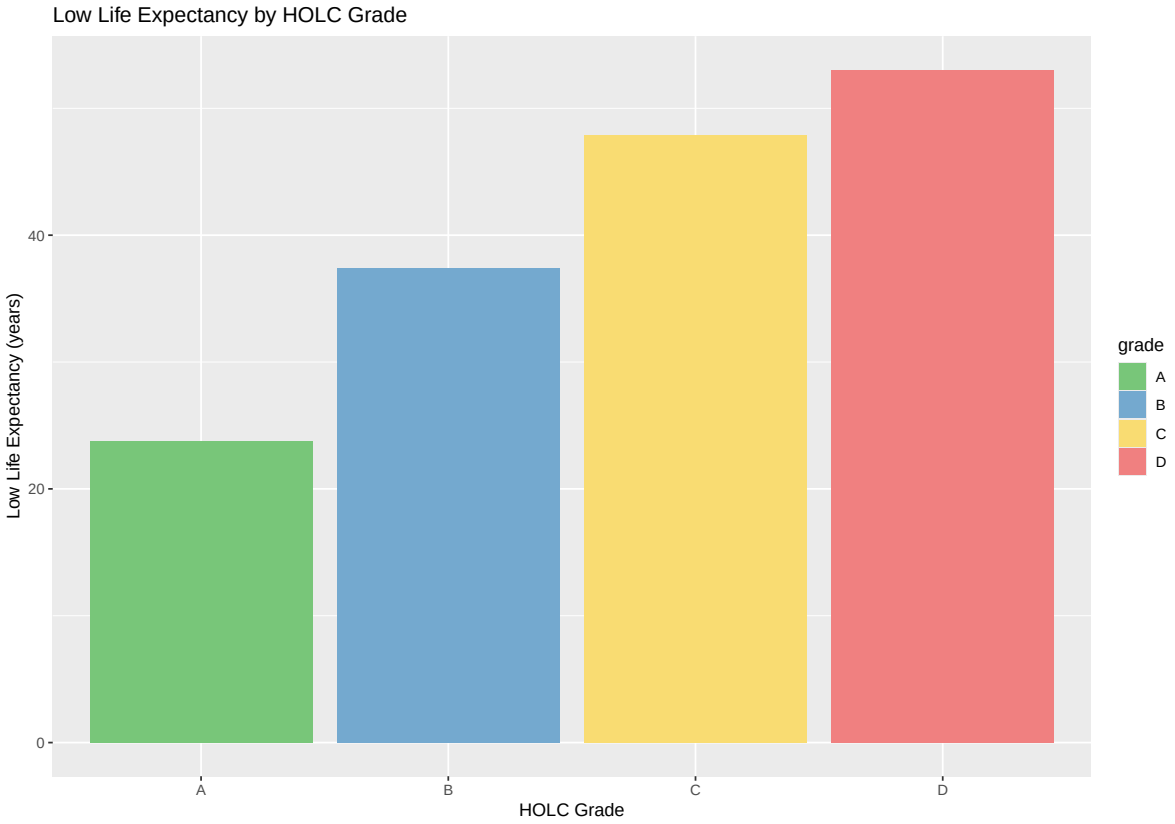
p1
```


% low income by HOLC Grade
Low Life Expectancy (years)
0.4
0.3
0.2
0.1
0.0
A
B
C
D
HOLC Grade
grade
A
B
C
D

```
p2
```


Particulate matter 2.5 by HOLC Grade
Low Life Expectancy (years)
80
60
40
20
0
A
B
C
D
HOLC Grade
grade
A
B
C
D

```
p3
```



4. Write a brief paragraph reflecting on these results

- Interpret the patterns you observe in your results
- Discuss potential relationships between historical redlining grades and current environmental/socioeconomic conditions

LA county is heavily urbanized resulting in less greenery and hotter neighborhoods which leads to higher redlining grades. This is apparent in our dataset with more than half of our census block occupying grade C and D. Grade D consistently had the highest mean values among all the other grades. The graphs followed a descending order from grade D to A. These high mean of percentage of low income, low life expectancy, and particulate matter supports the validity of the claim that census tracts graded "D" experienced 2.6 times the rate of pedestrian deaths compared to "A"-graded areas.

Part 2: Legacy of redlining in pedstrian and bike accidents

1. A figure summarizing the percent of observations within redlined neighborhoods within each HOLC grade. Create a visualizations that shows:

- The percentage of bike and pedestrian crashes and fatalities within each HOLC grade
- Include an appropiate title, axis labels, and legend
- Hints: Make sure to filter for bike and pedestrian crahes(PEDESTRIAN_ACCIDENT and BICYCLE_ACCIDENT). Look at the difference in NAs between POINT_X/POINT_Y and LONGITUDE/LATITUDE before you decide which to use. Ensure the SWITRS data and HOLC dataset have matching CRS', then perform a spatial join to assign each crash to a corresponding HOLC grade.

Data Setup

```
# change crash dataset from a dataframe to a sf spatial object
crash <- crash |>
  mutate(
      LONGITUDE = as.numeric(na_if(str_trim(LONGITUDE), "")),
      LATITUDE = as.numeric(na_if(str_trim(LATITUDE), ""))
  ) |>
  drop_na(LONGITUDE, LATITUDE) |>
  st_as_sf(
          coords = c('LONGITUDE', 'LATITUDE'),
          crs = st_crs(4326)
  )

crash <- st_transform(crash, st_crs(holc))
```

Coordinate Reference System Verification

```
if(st_crs(crash) == st_crs(los_angeles)) {
  print('crs matches')
} else {
  warning('crs do not match!')
}
```

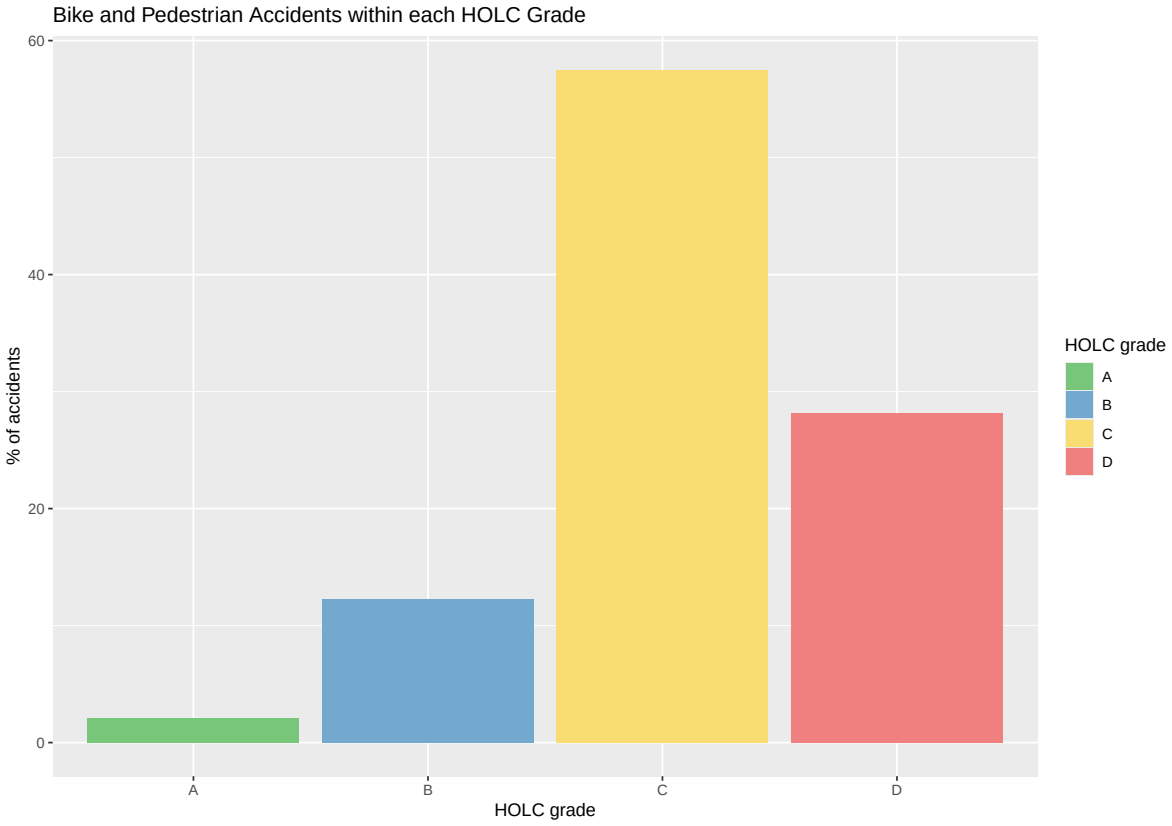
```
[1] "crs matches"
```

Join table and calculate percentage of accidents in each grade

```
# join crash and holc table
accidents <- st_join(crash, holc["grade"])

# remove NA values and calculate percentage of accidents in each grade
accidents <- accidents |>
  mutate(
    PEDESTRIAN_ACCIDENT = na_if(str_trim(PEDESTRIAN_ACCIDENT), ""),
    BICYCLE_ACCIDENT = na_if(str_trim(BICYCLE_ACCIDENT), "")
  ) |>
  drop_na(grade) |>
  group_by(grade) |>
  summarize(pa = sum(!is.na(PEDESTRIAN_ACCIDENT)),
            ba = sum(!is.na(BICYCLE_ACCIDENT)),
            total_accidents = ba + pa
  ) |>
  mutate(percentage_accident_fatalities = total_accidents / sum(total_accidents) * 100)
```

```
# create barplots comparing bike/pedestrian accidents across HOLC grades
ggplot(accidents, aes(x = grade)) +
  geom_col(aes(y = percentage_accident_fatalities, fill = c('A', 'B', 'C', 'D'))) +
  scale_fill_manual(values = c(
          "A" = "#78C679",
          "B" = "#74A9CF",
          "C" = "#F9DC72",
          "D" = "#F08080"
  )) +
  labs(
    x = "HOLC grade",
    y = "% of accidents",
    fill = "HOLC grade",
    title = "Bike and Pedestrian Accidents within each HOLC Grade")
```



Data Citation

Data	Citation	Link
Low Life Expectancy	U.S. Environmental Protection Agency (EPA). 2023. *EJScreen Version 2.2.*	EPA EJScreen
PM2.5	U.S. Environmental Protection Agency (EPA). 2023. *EJScreen Version 2.2.*	EPA EJScreen
Low Income	U.S. Environmental Protection Agency (EPA). 2023. *EJScreen Version 2.2.*	EPA EJScreen

Data	Citation	Link
HOLC Redlining	Nelson, R. K., Winling, L., Marciano, R., Connolly, N., et al. (n.d.). *Mapping Inequality: Redlining in New Deal America*. University of Richmond, Digital Scholarship Lab.	Mapping Inequality
Bicycle and Pedestrian Crashes	California Highway Patrol. (n.d.). *Statewide Integrated Traffic Records System (SWITRS)*.	Transportation Injury Mapping System
Los Angeles County	U.S. Environmental Protection Agency (EPA). 2023. *EJScreen Version 2.2*.	EPA EJScreen

Data sources used in the Los Angeles County to analyze the relationship between historical redlining and traffic crash analysis. The original EPA website no longer exists but the link provided is an alternative.